into energy (which would become the foundation for the atomic bomb; see Chapter 6), and the inertia of every object must increase so rapidly, as its speed approaches the speed of light, that no matter how hard one pushes on the object, one can never make it reach or surpass the speed of light ("nothing can go faster than light").[4]

In late June, Einstein wrote a technical article describing his ideas and their consequences, and mailed it off to the *Annalen der Physik.* His article carried the somewhat mundane title "On the Electrodynamics of Moving Bodies." But it was far from mundane. A quick perusal showed Einstein, the Swiss Patent Office's "technical expert third class," proposing a whole new foundation for physics, proposing a metaprinciple that all future physical laws must obey, radically revising our concepts of space and time, and deriving spectacular consequences. Einstein's new foundation and its consequences would soon come to be known as *special relativity* ("special" because it correctly describes the Universe only in those special situations where gravity is unimportant).

Einstein's article was received at the offices of the *Annalen der Physik* in Leipzig on 30 June 1905. It was perused for accuracy and importance by a referee, was passed as acceptable, and was published.

In the weeks after publication, Einstein waited expectantly for a response from the great physicists of the day. His viewpoint and conclusions were so radical and had so little experimental basis that he expected sharp criticism and controversy. Instead, he was met with stony silence. Finally, many weeks later, there arrived a letter from Berlin: Max Planck wanted clarification of some technical issues in the paper. Einstein was overjoyed! To have the attention of Planck, one of the most renowned of all living physicists, was deeply satisfying. And when Planck went on, the following year, to use Einstein's principle of relativity as a central tool in his own research, Einstein was further heartened. Planck's approval, the gradual approval of other leading physicists, and most important his own supreme self-confidence held Einstein firm throughout the following twenty years as the controversy he had expected did, indeed, swirl around his relativity theory. The controversy was still so strong in 1922 that, when the secretary of the Swedish Academy of Sciences informed Einstein by telegram that he had won the Nobel Prize, the telegram stated explicitly that relativity was *not* among the works on which the award was based.